

# SHIVANSH SINGH

## Senior Data Engineer | Cloud Data Platforms

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### PROFESSIONAL SUMMARY

Senior Data Engineer with 10+ years architecting enterprise data platforms and ETL/ELT pipelines on AWS at national scale (50 states). Drove \$360,000+ in annual cloud cost savings and 83% pipeline latency reduction. Partners with analysts and product teams to translate complex business requirements into reliable, production-grade data platforms. Currently pursuing Microsoft Fabric DP-600 certification.

### CORE COMPETENCIES

**Cloud & Platforms:** AWS (Lambda, EMR, RDS/Aurora, S3, DynamoDB, Kinesis, Athena, SQS, SNS)

**Data Architecture:** Layered Pipeline Architecture · ETL/ELT Pipelines · Streaming & Batch Processing · Late-Arriving Data & DLQ Reprocessing · Schema Change Management · Multi-Source Data Integration

**Data Engineering:** Apache Spark · Apache Kafka · Apache Airflow · Elasticsearch · Data Quality Frameworks · Data Governance · Pipeline Observability · High-Volume Reprocessing

**Languages & Tools:** Python · SQL · Java · Terraform · Docker · GitHub Actions · CI/CD · JIRA · Git

**Leadership:** Technical SME · Cross-functional Stakeholder Management · Engineering Mentorship · Solution Architecture · Agile/Scrum · Technical Documentation

### PROFESSIONAL EXPERIENCE

**ROCKET COMPANIES (Rocket Innovation Studio & Rocket Homes) | Detroit, MI / Toronto (Remote)**

*Data Engineer | Nov 2018 - Jul 2025*

Built and maintained the enterprise data platform powering Rocket Homes' national real estate expansion.

- Designed and operated a two-layer pipeline architecture (Raw + Conformed) across all 50 US states, integrating structured exports, API payloads, event streams, and third-party feeds into a unified data platform.
- Architected Spark-based ETL/ELT pipelines processing millions of records across distributed EMR clusters with partition pruning, broadcast joins, and predicate pushdown, reducing pipeline latency 83% and maintenance overhead 30%.
- Consolidated 50 state-dedicated EMR clusters into 8 shared clusters (\$300K savings) and migrated Aurora PostgreSQL to Serverless v2 (\$60K savings), delivering \$360K+ in annual cloud cost reduction.
- Built enterprise data quality frameworks with schema enforcement, business-rule validation, dead-letter queuing, and dry-run validation for 1M+ record reprocessing jobs, eliminating silent corruption risk.
- Built idempotent reprocessing pipelines for late-arriving data and Kafka DLQ recovery, including bulk reprocessing of 350,000+ records to directly enable a consumer product launch.
- Managed schema evolution in live production pipelines, adding enrichment and computed fields to conformed data structures while maintaining backward compatibility for existing consumers.
- Co-led enterprise CloudFormation-to-Terraform migration, designed reusable module library, and implemented CI/CD via GitHub Actions with deployment in under 10 minutes. Served as Terraform SME.
- Executed enterprise GitHub to GitHub EMU security migration maintaining 100% platform uptime with zero data loss.
- Directed Apache Airflow v2.8.1 upgrade with Docker-based local debugging, increasing developer velocity 25%.

- Owned Elasticsearch indices powering full-text search across millions of records with geo-bounding-box queries and faceted filtering. Managed DynamoDB pipeline state store with upsert patterns.
- Served as designated SME for PostgreSQL and Terraform migrations. Mentored junior engineers, authored onboarding documentation, and facilitated cross-functional alignment across Product, Analytics, Infrastructure, and Security teams.

## **AMROCK (formerly Title Source) | Detroit, MI**

*Software Engineer - Machine Learning | Feb 2016 - Nov 2018*

- Built the company's first ML pipeline for mortgage document classification: K-Means clustering for automated training data preparation (saving 160+ hours per cycle), Naive Bayes classifier achieving 97% classification rate and 96% accuracy at scale.
- Architected Flask microservices for production model serving with versioning, latency SLAs, and failure handling. First model-to-production deployment in company history.

## **CERTIFICATIONS & PROFESSIONAL DEVELOPMENT**

- **Microsoft Fabric Analytics Engineer Associate (DP-600)** - In Progress (target: Q2 2026)
- **AWS Certified Solutions Architect** - Cloud infrastructure expertise (background credential)

## **SIDE PROJECTS & INNOVATION**

**DriveBookAI - Multi-Tenant SaaS Booking Platform** Production-ready booking platform for driving schools. Multi-tenant PostgreSQL with RLS, stateless Node.js/Fastify API with Redis session externalization, embeddable JS widget, and AI integration layer (Claude API, Whisper).

**SPX Options Analytics Platform** Platform for analyzing SPX options pricing and greeks. Multi-source ETL pipeline (IBKR TWS, Polygon.io), custom Black-Scholes greeks engine (first and second-order), and tiered PostgreSQL/Cloudflare R2 Parquet storage.

**FuturesEdgeTrader - Multi-Agent Algorithmic Trading System** Algorithmic trading system for E-mini futures. Event-driven 7-agent Claude-powered pipeline, multi-broker execution (Rithmic, Tradovate, IBKR), dual-path PostgreSQL/Supabase data layer, and Telegram/Streamlit monitoring.

**AI Real Estate Buyer Report (Hackathon)** AI-powered tool for generating comprehensive real estate buyer reports. Multi-agent system with specialized data-gathering agents and a compiler agent synthesizing outputs into structured PDF reports. RAG pipeline with pgvector for semantic retrieval.

## **EDUCATION**

- **M.S. Computer Science, 2015** | University of Florida, Gainesville, FL | GPA: 3.8/4.0
- **B.Tech Computer Science, 2014** | IIIT, Noida, India | GPA: 8.5/10.0